National Testing Agency

Question Paper Name :	B Tech 2 Apr 2025 Shift 2
Subject Name :	B. Tech
Duration :	180
Total Marks :	300
Display Marks:	Yes

B. Tech

Group Number :	1
Group Maximum Duration :	0
Group Minimum Duration :	180
Show Attended Group? :	No
Edit Attended Group? :	No
Break time :	0
Group Marks :	300

Mathematics Section A

Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 1 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the mean and the variance of $6,4,a,8,b,12,10,13$ are 9 and 9.25 respectively, then $a+b+ab$ is equal to :

Options :

A. 103

B. 106

C. 100

D. 105

Question Number : 2 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $f:[1,\infty)\to[2,\infty)$ be a differentiable function. If $10\int_1^x f(\mathrm{t})\mathrm{dt}=5xf(x)-x^5-9$ for all $x\geqslant 1$, then the value of $f(3)$ is :

Options :

A. 22

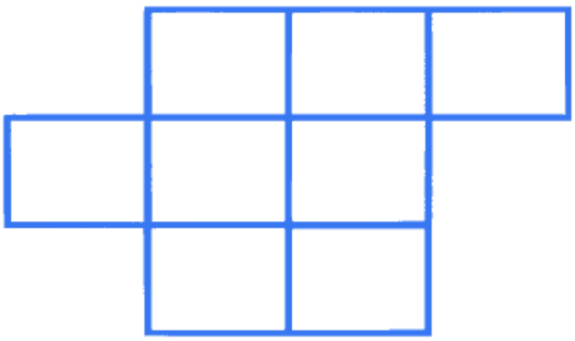
B. 26

C. 32

D. 18

Question Number : 3 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of ways, in which the letters A, B, C, D, E can be placed in the 8 boxes of the figure below so that no row remains empty and at most one letter can be placed in a box, is :



Options :

A. 5880

B. 840

C. 960

D. 5760

Question Number : 4 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the point P of the focal chord PQ of the parabola $y^2 = 16x$ be $(1, -4)$. If the focus of the parabola divides the chord PQ in the ratio $m : n, \gcd(m, n) = 1$, then $m^2 + n^2$ is equal to :

Options :

A. 17

B. 37

C. 10

D. 26

Question Number : 5 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the system of equations

$$2x + \lambda y + 3z = 5$$
$$3x + 2y - z = 7$$
$$4x + 5y + \mu z = 9$$

has infinitely many solutions, then $\left(\lambda^2 + \mu^2\right)$ is equal to :

Options :

A. 30

B. 26

C. 22

D. 18

Question Number : 6 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let A be a 3×3 real matrix such that $A^2(A - 2I) - 4(A - I) = O$, where I and O are the identity and null matrices, respectively. If $A^5 = \alpha A^2 + \beta A + \gamma I$, where α, β, and γ are real constants, then $\alpha + \beta + \gamma$ is equal to :

Options :

A. 76

B. 12

C. 4

D. 20

Question Number : 7 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

$$If \lim_{x \to 0} \frac{\cos(2x) + a\cos(4x) - b}{x^4} is\ finite,\ then\ (a + b)\ is\ equal\ to :$$

Options :

A. 0

B. $\frac{3}{4}$

C. -1

D. $\frac{1}{2}$

Question Number : 8 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the image of the point P(1,0,3) in the line joining the points A(4,7,1) and B(3,5,3) is $Q(\alpha, \beta, \gamma)$, then $\alpha + \beta + \gamma$ is equal to :

Options :

A. $\frac{46}{3}$

B. 18

C. 13

D. $\frac{47}{3}$

Question Number : 9 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If $\theta \epsilon \left[-\frac{7\pi}{6}, \frac{4\pi}{3}\right]$, then the number of solutions of $\sqrt{3}\ \text{cosec}^2\theta - 2(\sqrt{3} - 1)\text{cosec}\,\theta - 4 = 0$, is equal to :

Options :

A. 7

B. 10

C. 6

D. 8

Question Number : 10 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $A = \{1,2,3, \ldots ., 100\}$ and R be a relation on A such that $R = \{(a, b) : a = 2b + 1\}$. Let $(a_1, a_2), (a_2, a_3), (a_3, a_4), \ldots ., (a_k, a_{k+1})$ be a sequence of k elements of R such that the second entry of an ordered pair is equal to the first entry of the next ordered pair. Then the largest integer k , for which such a sequence exists, is equal to :

Options :

A. 6

B. 8

C. 7

D. 5

Question Number : 11 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the domain of the function $f(x) = \frac{1}{\sqrt{10+3x-x^2}} + \frac{1}{\sqrt{x+|x|}}$ is (a, b), then $(1+a)^2 + b^2$ is equal to :

Options :

A. 29

B. 30

C. 25

D. 26

Question Number : 12 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let the area of the triangle formed by a straight line $L: x + by + c = 0$ with co-ordinate axes be 48 square units. If the perpendicular drawn from the origin to the line L makes an angle of 45° with the positive x-axis, then the value of $b^2 + c^2$ is :

Options :

A. 90

B. 83

C. 93

D. 97

Question Number : 13 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The line L_1 is parallel to the vector $\vec{a} = -3\hat{i} + 2\hat{j} + 4\hat{k}$ and passes through the point (7,6,2) and the line L_2 is parallel to the vector $\vec{b} = 2\hat{i} + \hat{j} + 3\hat{k}$ and passes through the point (5,3,4). The shortest distance between the lines L_1 and L_2 is :

Options :

A. $\frac{23}{\sqrt{38}}$

B. $\frac{21}{\sqrt{38}}$

C. $\frac{23}{\sqrt{57}}$

D. $\frac{21}{\sqrt{57}}$

Question Number : 14 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The number of terms of an A.P. is even; the sum of all the odd terms is 24 , the sum of all the even terms is 30 and the last term exceeds the first by $\frac{21}{2}$. Then the number of terms which are integers in the A.P. is :

Options :

A. 6

B. 4

C. 8

D. 10

Question Number : 15 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let (a, b) be the point of intersection of the curve $x^2 = 2y$ and the straight line $y - 2x - 6 = 0$ in the second quadrant. Then the integral $\mathrm{I} = \int_{\mathrm{a}}^{\mathrm{b}} \frac{9x^2}{1+5^x}\,\mathrm{d}x$ is equal to :

Options :

A. 27

B. 18

C. 24

D. 21

Question Number : 16 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If the length of the minor axis of an ellipse is equal to one fourth of the distance between the foci, then the eccentricity of the ellipse is :

Options :

A. $\frac{3}{\sqrt{19}}$

B. $\frac{\sqrt{3}}{16}$

C. $\frac{4}{\sqrt{17}}$

D. $\frac{\sqrt{5}}{7}$

Question Number : 17 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

$$If \sum_{r=0}^{10} \left(\frac{10^{r+1}-1}{10^r}\right).{}^{11}C_{r+1} = \frac{\alpha 11 - 11^{11}}{10^{10}}, \text{ then } \alpha \text{ is equal to:}$$

Options :

A. 11

B. 20

C. 24

D. 15

Question Number : 18 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

$4\int_0^1 \left(\frac{1}{\sqrt{3+x^2}+\sqrt{1+x^2}}\right)dx - 3\log_e(\sqrt{3}\,)$ is equal to :

Options :

A. $2-\sqrt{2}\,-\log_e(1+\sqrt{2}\,)$

B. $2+\sqrt{2}\,+\log_e(1+\sqrt{2}\,)$

C. $2+\sqrt{2}\,-\log_e(1+\sqrt{2}\,)$

D. $2-\sqrt{2}\,+\log_e(1+\sqrt{2}\,)$

Question Number : 19 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Let $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} + 2\hat{j} + 5\hat{k}$ and a vector $\vec{c}$ be such that $(\vec{a} - \vec{c}) \times \vec{b} = -18\hat{i} - 3\hat{j} + 12\hat{k}$ and $\vec{a} \cdot \vec{c} = 3$. If $\vec{b} \times \vec{c} = \vec{d}$, then $|\vec{a} \cdot \vec{d}|$ is equal to :

Options :

A. 15

B. 18

C. 12

D. 9

Question Number : 20 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given three indentical bags each containing 10 balls, whose colours are as follows :

	Red	Blue	Green
Bag I	3	2	5
Bag II	4	3	3
Bag III	5	1	4

A person chooses a bag at random and takes out a ball. If the ball is Red, the probability that it is from bag I is p and if the ball is Green, the probability that it is from bag III is q, then the value of $\left(\frac{1}{p}+\frac{1}{q}\right)$ is:

Options :

A. 6

B. 9

C. 7

D. 8

Mathematics Section B

Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 21 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If $y=\cos\left(\frac{\pi}{3}+\cos^{-1}\frac{x}{2}\right)$, then $(x-y)^2+3y^2$ is equal to

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 22 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If the set of all $a \in \mathbf{R}-\{1\}$, for which the roots of the equation $(1-a)x^2+2(a-3)x+9=0$ are positive is $(-\infty,-\alpha]\cup[\beta,\gamma)$, then $2\alpha+\beta+\gamma$ is equal to .

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 23 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

If the sum of the first 10 terms of the series $\frac{4\cdot1}{1+4\cdot1^4}+\frac{4\cdot2}{1+4\cdot2^4}+\frac{4\cdot3}{1+4\cdot3^4}+$ ... .. is $\frac{m}{n}$, where $\gcd(m,n)=1$, then $m+n$ is equal to _____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 24 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let $y=y(x)$ be the solution of the differential equation $\frac{dy}{dx}+2y\sec^2 x=2\sec^2 x+3\tan x\cdot\sec^2 x$ such that $y(0)=\frac{5}{4}$. Then $12\left(y\left(\frac{\pi}{4}\right)-e^{-2}\right)$ is equal to_____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 25 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

Let A(4, −2), B(1,1) and C(9, −3) be the vertices of a triangle ABC . Then the maximum area of the parallelogram AFDE, formed with vertices D, E and F on the sides BC, CA and *AB* of the triangle *ABC* respectively, is_____

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Physics Section A

Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 26 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The moment of inertia of a circular ring of mass M and diameter r about a tangential axis lying in the plane of the ring is :

Options :

A. $\frac{3}{8}Mr^2$

B. $2Mr^2$

C. $\frac{1}{2}Mr^2$

D. $\frac{3}{2}Mr^2$

Question Number : 27 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In a moving coil galvanometer, two moving coils M_1 and M_2 have the following particulars :

$$R_1 = 5\Omega,\ N_1 = 15,\ A_1 = 3.6 \times 10^{-3}\ m^2,\ B_1 = 0.25\ T$$
$$R_2 = 7\Omega,\ N_2 = 21,\ A_2 = 1.8 \times 10^{-3}\ m^2,\ B_2 = 0.50\ T$$

Assuming that torsional constant of the springs are same for both coils, what will be the ratio of voltage sensitivity of M_1 and M_2 ?

Options :

A. 1 : 1

B. 1 : 3

C. 1 : 2

D. 1 : 4

Question Number : 28 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Net dipole moment of a polar linear isotropic dielectric substance is not zero even in the absence of an external electric field.

Reason (R) : In absence of an external electric field, the different permanent dipoles of a polar dielectric substance are oriented in random directions.

In the light of the above statements, choose the most appropriate answer from the options given below :

Options :

A. Both (A) and (R) are correct and (R) is the correct explanation of (A) 2.

B. Both (A) and (R) are correct but (R) is not the correct explanation of (A)

C. (A) is not correct but (R) is correct

D. (A) is correct but (R) is not correct

Question Number : 29 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

If μ_0 and ϵ_0 are the permeability and permittivity of free space, respectively, then the dimension of $\left(\frac{1}{\mu_0\epsilon_0}\right)$ is :

Options :

A. T^2/L

B. L^2/T^2

C. T^2/L^2

D. L/T^2

Question Number : 30 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A sinusoidal wave of wavelength 7.5 cm travels a distance of 1.2 cm along the x-direction in 0.3 sec . The crest P is at $x = 0$ at t = 0sec and maximum displacement of the wave is 2 cm . Which equation correctly represents this wave?

Options :

A. $y = 2\cos(0.83x - 3.35t)\text{cm}$

B. $y = 2\sin(0.83x - 3.5\text{t})\text{cm}$

C. $y = 2\cos(0.13x - 0.5t)\text{cm}$

D. $y = 2\cos(3.35x - 0.83\text{t})\text{cm}$

Question Number : 31 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A solenoid having area A and length ' l ' is filled with a material having relative permeability 2. The magnetic energy stored in the solenoid is :

Options :

A. $\frac{B^2\,Al}{4\mu_0}$

B. $\frac{B^2\,Al}{2\mu_0}$

C. $\frac{B^2\,Al}{\mu_0}$

D. B^2Al

Question Number : 32 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two identical objects are placed in front of convex mirror and concave mirror having same radii of curvature of 12 cm , at same distance of 18 cm from the respective mirrors. The ratio of sizes of the images formed by convex mirror and by concave mirror is :

Options :

A. 2

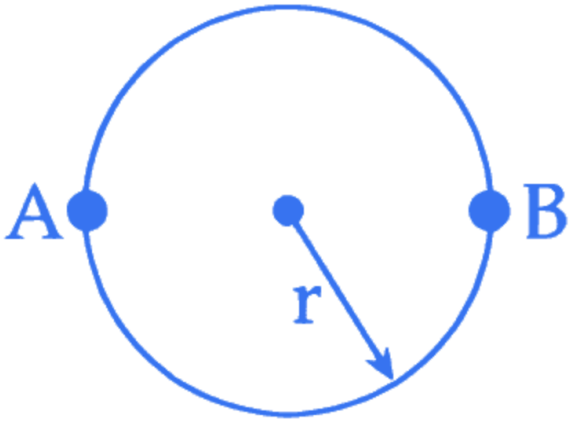
B. $1/2$

C. $1/3$

D. 3

Question Number : 33 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A sportsman runs around a circular track of radius r such that he traverses the path $ABAB$. The distance travelled and displacement, respectively, are



Options :

A. $\pi r, 3r$

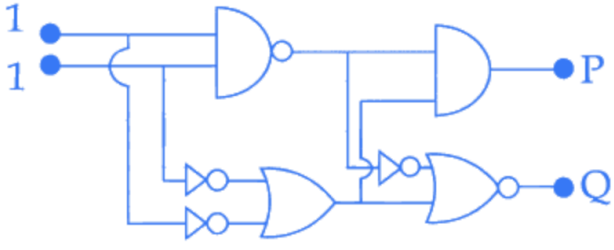
B. $2r, 3\pi r$

C. $3\pi r, 2r$

D. $3\pi r, \pi r$

Question Number : 34 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In the digital circuit shown in the figure, for the given inputs the P and Q values are :



Options :

A. $P = 0, Q = 1$

B. $P = 1, Q = 0$

C. $P = 0, Q = 0$

D. $P = 1, Q = 1$

Question Number : 35 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

An electron with mass ' m ' with an initial velocity $(t = 0)\vec{v} = v_0\hat{i}\,(v_0 > 0)$ enters a magnetic field $\vec{B} = B_0\hat{j}$. If the initial de-Broglie wavelength at $t = 0$ is λ_0 then its value after time ' t ' would be :

Options :

A. $\frac{\lambda_0}{\sqrt{1 - \frac{e^2 B_0^2 t^2}{m^2}}}$

B. λ_0

C. $\lambda_0\sqrt{1 + \frac{e^2 B_0^2 t^2}{m^2}}$

D. $\frac{\lambda_0}{\sqrt{1 + \frac{e^2 B_0^2 t^2}{m^2}}}$

Question Number : 36 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A bi-convex lens has radius of curvature of both the surfaces same as 1/6 cm. If this lens is required to be replaced by another convex lens having different radii of curvatures on both sides $(R_1 \neq R_2)$, without any change in lens power then possible combination of R_1 and R_2 is :

Options :

A. $\frac{1}{3}$ cm and $\frac{1}{7}$ cm

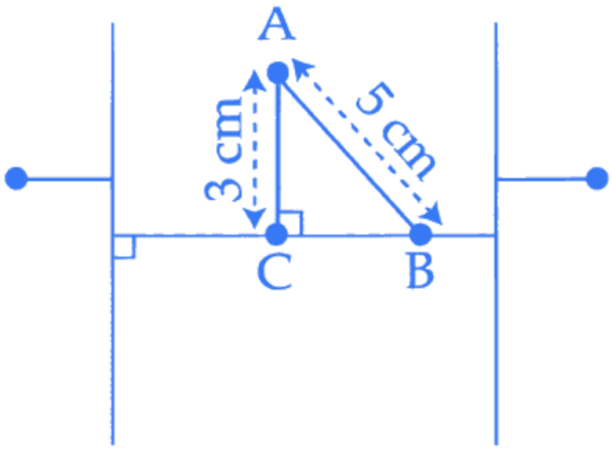
B. $\frac{1}{5}$ cm and $\frac{1}{7}$ cm

C. $\frac{1}{3}$ cm and $\frac{1}{3}$ cm

D. $\frac{1}{6}$ cm and $\frac{1}{9}$ cm

Question Number : 37 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two large plane parallel conducting plates are kept 10 cm apart as shown in figure. The potential difference between them is V . The potential difference between the points A and *B* (shown in the figure) is :



Options :

A. $1\ V$

B. $\frac{1}{4}V$

C. $\frac{3}{4}V$

D. $\frac{2}{5}V$

Question Number : 38 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Two water drops each of radius ' *r* ' coalesce to form a bigger drop. If ' *T* ' is the surface tension, the surface energy released in this process is :

Options :

A. $4\pi r^2\ T\left[2-2^{1/3}\right]$

B. $4\pi r^2\ T[1+\sqrt{2}\ ]$

C. $4\pi r^2\ T\left[2-2^{2/3}\right]$

D. $4\pi r^2\ T[\sqrt{2}\ -1]$

Question Number : 39 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Energy released when two deuterons $\left({}_1H^2\right)$ fuse to form a helium nucleus $\left({}_2He^4\right)$ is : (Given : Binding energy per nucleon of ${}_1H^2 = 1.1MeV$ and binding energy per nucleon of ${}_2He^4 = 7.0MeV$)

Options :

A. 26.8 MeV

B. 8.1 MeV

C. 23.6 MeV

D. 5.9 MeV

Question Number : 40 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match List - I with List - II.

	List - I	*List – II*
(A)	Heat capacity of body	(I) J kg^{-1}
(B)	Specific heat capacity of body	(II) J K^{-1}
(C)	Latent heat	(III) J kg^{-1} K^{-1}
(D)	Thermal conductivity	(IV) J m^{-1} K^{-1} s^{-1}

Choose the correct answer from the options given below :

Options :

A. (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

B. (A)-(III), (B)-(I), (C)-(II), (D)-(IV)

C. (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

D. (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Question Number : 41 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider a circular loop that is uniformly charged and has a radius $a\sqrt{2}$. Find the position along the positive z-axis of the cartesian coordinate system where the electric field is maximum if the ring was assumed to be placed in xy plane at the origin :

Options :

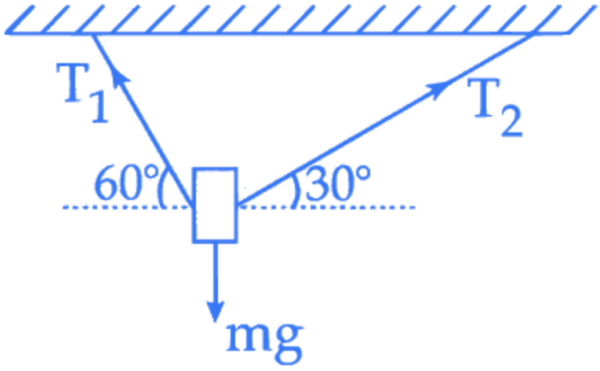
A. a

B. $a/2$

C. 0

D. $a/\sqrt{2}$

Question Number : 42 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical



A body of mass 1 kg is suspended with the help of two strings making angles as shown in figure. Magnitudes of tensions T_1 and T_2, respectively, are (in N) :

(Take acceleration due to gravity 10 m/s^2)

Options :

A. $5\sqrt{3}$, 5

B. $5\sqrt{3}$, $5\sqrt{3}$

C. 5,5

D. $5,5\sqrt{3}$

Question Number : 43 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Identify the characteristics of an adiabatic process in a monoatomic gas. (A) Internal energy is constant. (B) Work done in the process is equal to the change in internal energy. (C) The product of temperature and volume is a constant. (D) The product of pressure and volume is a constant. (E) The work done to change the temperature from T_1 to T_2 is proportional to $(T_2 - T_1)$. Choose the correct answer from the options given below :

Options :

A. (B), (D) only

B. (B), (E) only

C. (A), (C), (E) only

D. (A), (C), (D) only

Question Number : 44 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given a charge q , current I and permeability of vacuum μ_{o^*}. Which of the following quantity has the dimension of momentum ?

Options :

A. qI/μ_o

B. $q^2\mu_o I$

C. $q\mu_0/I$

D. $q\mu_o I$

Question Number : 45 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Assuming the validity of Bohr's atomic model for hydrogen like ions the radius of Li^{++} ion in its ground state is given by $\frac{1}{X}a_0$, where $X =$ _____ (Where a_0 is the first Bohr's radius.)

Options :

A. 2

B. 9

C. 1

D. 3

Physics Section B

Section Number :	4
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 46 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A ray of light suffers minimum deviation when incident on a prism having angle of the prism equal to 60°. The refractive index of the prism material is $\sqrt{2}$. The angle of incidence (in degrees) is_____

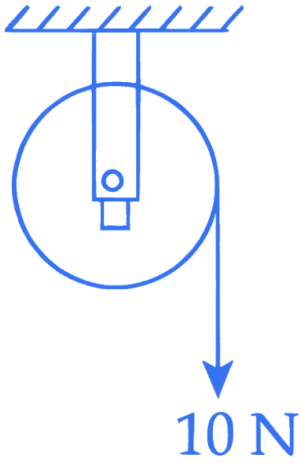
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 47 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The length of a light string is 1.4 m when the tension on it is 5 N . If the tension increases to 7 N , the length of the string is 1.56 m . The original length of the string is_____m.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 48 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript



A wheel of radius 0.2 m rotates freely about its center when a string that is wrapped over its rim is pulled by force of 10 N as shown in figure. The established torque produces an angular acceleration of $2 rad/s^2$. Moment of intertia of the wheel is_____ kg m^2. (Acceleration due to gravity $= 10\ m/s^2$)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 49 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The internal energy of air in $4\ m \times 4\ m \times 3\ m$ sized room at 1 atmospheric pressure will be_____ $\times 10^6$ J

(Consider air as diatomic molecule)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 50 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

A satellite of mass 1000 kg is launched to revolve around the earth in an orbit at a height of 270 km from the earth's surface. Kinetic energy of the satellite in this orbit is_____ $\times 10^{10}$ J.

(Mass of earth $= 6 \times 10^{24}$ kg, Radius of earth $= 6.4 \times 10^6$ m, Gravitational constant $= 6.67 \times 10^{-11} Nm^2\ kg^{-2}$)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Chemistry Section A

Section Number :	5
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	20
Number of Questions to be attempted :	20
Section Marks :	80
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 51 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Given below are two statements :

Statement (I) : Neopentane forms only one monosubstituted derivative.

Statement (II) : Melting point of neopentane is higher than n-pentane.

In the light of the above statements, choose the most appropriate answer from the options given below :

Options :

A. Statement I is incorrect but Statement II is correct

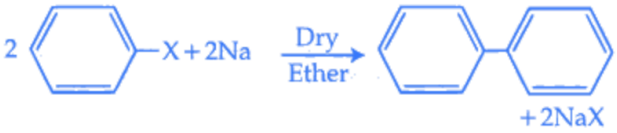
B. Both Statement I and Statement II are correct

C. Statement I is correct but Statement II is incorrect

D. Both Statement I and Statement II are incorrect

Question Number : 52 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match List - I with List - II.

List - I (Reaction)		**List - II (Name of reaction)**	
(A)		(I)	Lucas reaction
(B)	$ArN_2^+X^- \xrightarrow[HCl]{Cu} ArCl + N_2 \uparrow + CuX$	(II)	Finkelstein reaction
(C)	$C_2H_5Br + NaI \xrightarrow[Acetone]{Dry} C_2H_5I + NaBr$	(III)	Fittig reaction
(D)	$CH_3C(OH)(CH_3)CH_3 \xrightarrow[ZnCl_2]{HCl} CH_3C(Cl)(CH_3)CH_3$	(IV)	Gatterman reaction

Choose the correct answer from the options given below :

Options :

A. (A)-(III), (B)-(II), (C)-(IV), (D)-(I)

B. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

C. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

D. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Question Number : 53 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Electronic configuration of four elements A, B, C and D are given below :

(A) $1s^2 2s^2 2p^3$

(B) $1s^2 2s^2 2p^4$

(C) $1s^2 2s^2 2p^5$

(D) $1s^2 2s^2 2p^2$

Which of the following is the correct order of increasing electronegativity (Pauling's scale)?

Options :

A. D < A < B < C

B. $A < B < C < D$

C. $A < C < B < D$

D. $A < D < B < C$

Question Number : 54 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

A tetrapeptide, " x " on complete hydrolysis produced glycine (Gly), alanine (Ala), valine (Val), leucine (Leu) in equimolar proportion each. The number of tetrapeptides (sequences) possible involving each of these amino acids is :

Options :

A. 32

B. 24

C. 8

D. 16

Question Number : 55 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which among the following molecules is (a) involved in sp^3 d hybridization, (b) has different bond lengths and (c) has lone pair of electrons on the central atom?

Options :

A. XeF_4

B. XeF_2

C. PF_5

D. SF_4

Question Number : 56 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In 3,3-dimethylhex-1-en-4-yne, there are _____sp^3,_____sp^2 and_____ sp hybridised carbon atoms respectively.

Options :

A. 4,2,2

B. 2,4,2

C. 3,3,2

D. 2,2,4

Question Number : 57 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

In Dumas' method for estimation of nitrogen, 0.5 gram of an organic compound gave 60 mL of nitrogen collected at 300 K temperature and 715 mm Hg pressure. The percentage composition of nitrogen in the compound (Aqueous tension at 300 K = 15 mmHg) is_____%.

Options :

A. 20.87

B. 12.57

C. 1.257

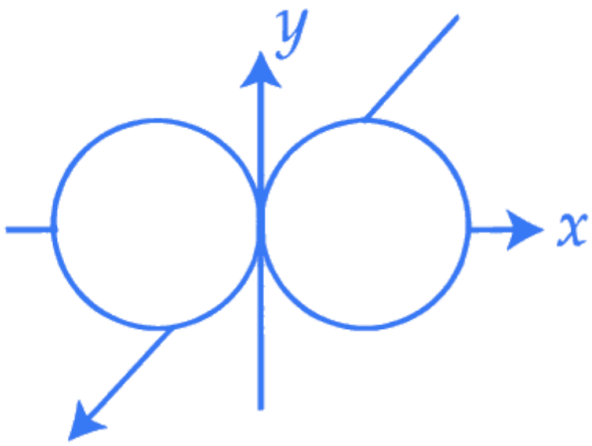
D. 18.67

Question Number : 58 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Which of the following statements are true?

(A) The subsidiary quantum number l describes the shape of the orbital occupied by the electron.

(B)



is the boundary surface diagram of the $2p_x$ orbital.

(C) The + and - signs in the wave function of the $2p_x$ orbital refer to charge.

(D) The wave function of $2p_x$ orbital is zero everywhere in the xy plane.

Choose the correct answer from the options given below :

Options :

A. (A) and (B) only

B. (B) and (D) only

C. (C) and (D) only

D. (A), (B) and (C) only

Question Number : 59 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Arrange the following in order of magnitude of work done by the system/on the system at constant temperature.

(a) $|w_{\text{reversible}}|$ for expansion in infinite stages.

(b) $|w_{\text{irreversible}}|$ for expansion in single stage.

(c) $|w_{\text{reversible}}|$ for compression in infinite stages.

(d) $|w_{\text{irreversible}}|$ for compression in single stage.

Choose the correct answer from the options given below :

Options :

A. $d > c = a > b$

B. $c = a > d > b$

C. $a > c > b > d$

D. $a > b > c > d$

Question Number : 60 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

When a concentrated solution of sulphanilic acid and 1-naphthylamine is treated with nitrous acid (273 K) and acidified with acetic acid, the mass (g) of 0.1 mole of product formed is :

(Given molar mass in gmol^{-1} H : 1, C : 12, N : 14, O : 16, S : 32)

Options :

A. 330

B. 33

C. 343

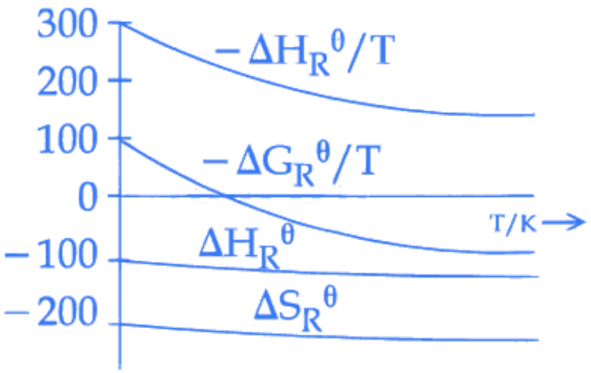
D. 66

Question Number : 61 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

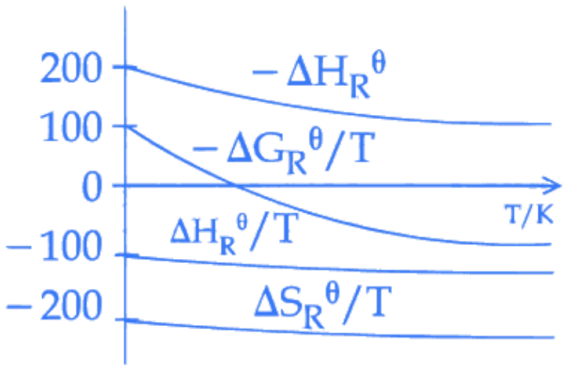
Which of the following graphs correctly represents the variation of thermodynamic properties of Haber's process?

Options :

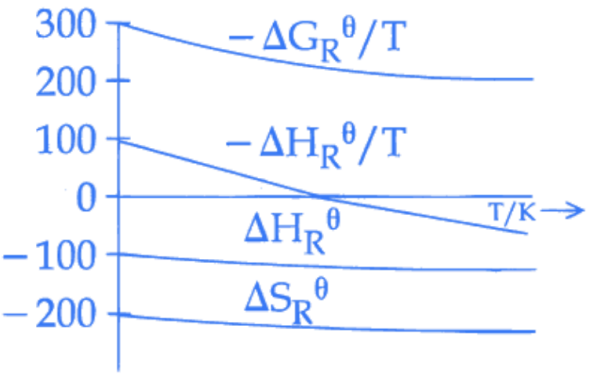
A.



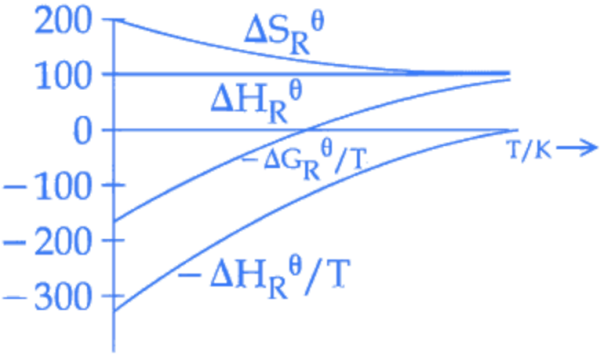
B.



C.



D.



Question Number : 62 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Reactant A converts to product D through the given mechanism (with the net evolution of heat):

$A \rightarrow B$ slow; ΔH = +ve

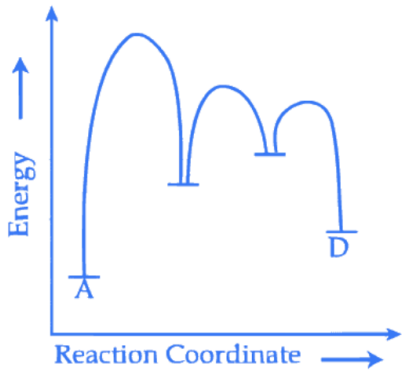
$B \rightarrow C$ fast; ΔH = -ve

$C \rightarrow D$ fast; ΔH = -ve

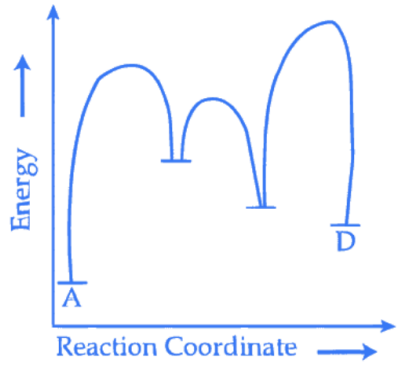
Which of the following represents the above reaction mechanism?

Options :

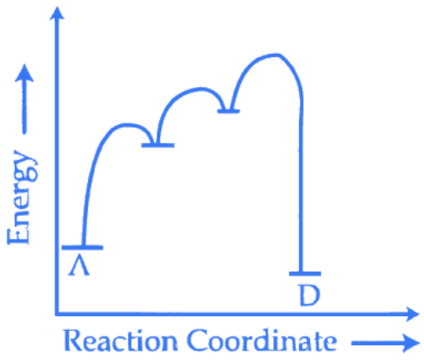
A.



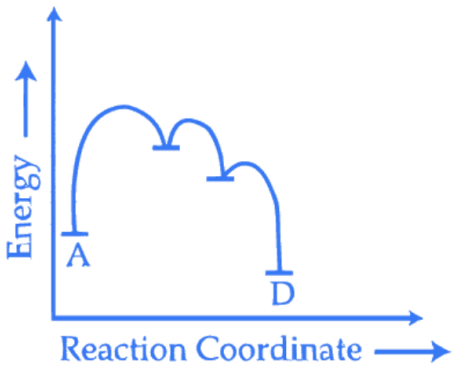
B.



C.



D.



Question Number : 63 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Formation of $Na_4\left[Fe(CN)_5NOS\right]$, a purple coloured complex formed by addition of sodium nitroprusside in sodium carbonate extract of salt indicates the presence of :

Options :

A. Sulphide ion

B. Sulphite ion

C. Sulphate ion

D. Sodium ion

Question Number : 64 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following reactions. From these reactions which reaction will give carboxylic acid as a major product ?

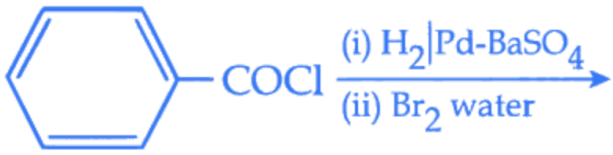
(A) $R-C\equiv N \xrightarrow[\text{mild condition}]{\text{(i) } \overset{+}{H}/H_2O}$

(B) $R-MgX \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } CO_2}$

(C) $R-C\equiv N \xrightarrow[\text{(ii) } H_3O^+]{\text{(i) } SnCl_2/HCl}$

(D) $R\cdot CH_2\cdot OH \xrightarrow{PCC}$

(E)



Choose the correct answer from the options given below :

Options :

A. A, B and E only

B. A and D only

C. B, C and E only

D. B and E only

Question Number : 65 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The nature of oxide $\left(TeO_2\right)$ and hydride $\left(TeH_2\right)$ formed by Te , respectively are :

Options :

A. Reducing and basic

B. Reducing and acidic

C. Oxidising and acidic

D. Oxidising and basic

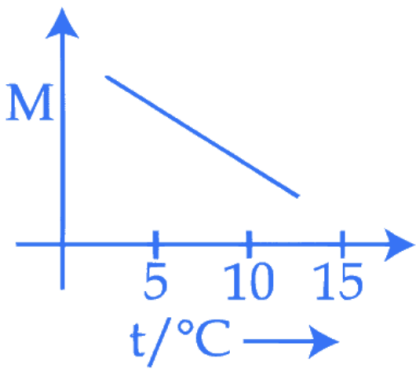
Question Number : 66 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

' x ' g of NaCl is added to water in a beaker with a lid. The temperature of the system is raised from 1°C to 25°C. Which out of the following plots, is best suited for the change in the molarity (M) of the solution with respect to temperature ?

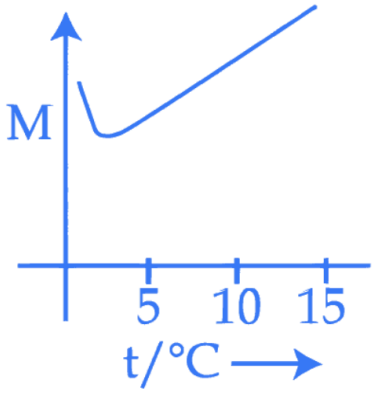
[Consider the solubility of NaCl remains unchanged over the temperature range]

Options :

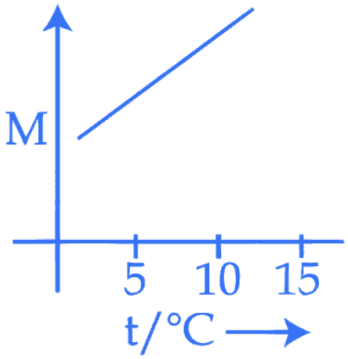
A.



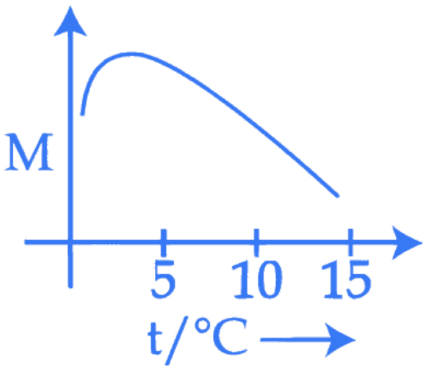
B.



C.



D.



Question Number : 67 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The type of hybridization and the magnetic property of $\left[MnCl_6\right]^{3-}$ are,

Options :

A. sp^3d^2, paramagnetic with four unpaired electrons.

B. d^2sp^3, paramagnetic with four unpaired electrons.

C. $sp^3\ d^2$, paramagnetic with two unpaired electrons.

D. d^2sp^3, paramagnetic with two unpaired electrons.

Question Number : 68 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Match List - I with List - II.

List - I (Purification technique)		**List - II (Mixture of organic compounds)**	
(A)	Distillation (simple)	(I)	Diesel + Petrol
(B)	Fractional distillation	(II)	Aniline + Water
(C)	Distillation under reduced pressure	(III)	Chloroform + Aniline
(D)	Steam distillation	(IV)	Glycerol + Spent-lye

Choose the correct answer from the options given below :

Options :

A. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

B. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

C. (A)-(II), (B)-(IV), (C)-(I), (D)-(III)

D. (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

Question Number : 69 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

The d-orbital electronic configuration of the complex among $\left[\mathrm{Co(en)_3}\right]^{3+}$, $\left[\mathrm{CoF_6}\right]^{3-}$, $\left[\mathrm{Mn(H_2O)_6}\right]^{2+}$ and $\left[\mathrm{Zn(H_2O)_6}\right]^{2+}$ that has the highest CFSE is :

Options :

A. $t_{2g}^{6}e_{g}^{0}$

B. $t_{2g}^{3}e_{g}^{2}$

C. $t_{2g}^{4}e_{g}^{2}$

D. $t_{2g}^{6}e_{g}^{4}$

Question Number : 70 Question Type : MCQ Option Shuffling : Yes Display Question Number : Yes Is Question Mandatory : No Single Line Question Option : No Option Orientation : Vertical

Consider the following chemical equilibrium of the gas phase reaction at a constant temperature :

$A(g) \rightleftharpoons B(g) + C(g)$

If p being the total pressure, K_p is the pressure equilibrium constant and α is the degree of dissociation, then which of the following is true at equilibrium?

Options :

A. If K_p value is extremely high compared to p, α becomes much less than unity

B. When p increases α increases

C. If p value is extremely high compared to $K_p, \alpha \approx 1$

D. When p increases α decreases

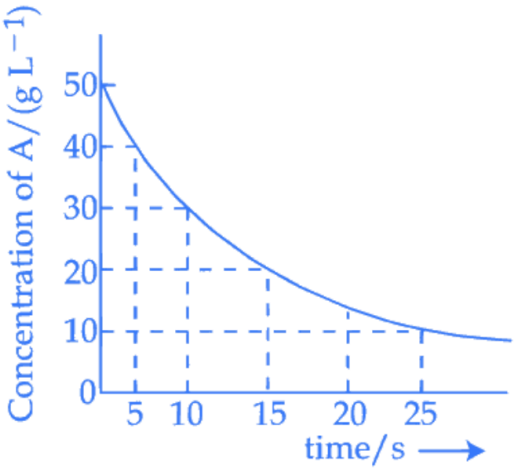
Chemistry Section B

Section Number :	6
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	5
Number of Questions to be attempted :	5
Section Marks :	20
Maximum Instruction Time :	0
Sub-Section Number :	1
Question Shuffling Allowed :	Yes
Is Section Default? :	No

Question Number : 71 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

For the reaction A → B the following graph was obtained. The time required (in seconds) for the concentration of A to reduce to 2.5 g L^{-1} (if the initial concentration of A was 50 g L^{-1}) is . (Nearest integer)

Given : $\log 2 = 0.3010$



Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes

Answers Type : Equal
Text Areas : PlainText

Question Number : 72 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

When 1 g each of compounds AB and AB_2 are dissolved in 15 g of water separately, they increased the boiling point of water by 2.7 K and 1.5 K respectively. The atomic mass of A (in *amu*) is_____ $\times 10^{-1}$ (Nearest integer)

(Given : Molal boiling point elevation constant is 0.5 K kg mol^{-1})

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 73 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

The spin-only magnetic moment value of M^{n+} ion formed among Ni, Zn, Mn and Cu that has the least enthalpy of atomisation is_____ . (in nearest integer) Here n is equal to the number of diamagnetic complexes among $K_2[NiCl_4]$, $[Zn(H_2O)_6]Cl_2$, $K_3[Mn(CN)_6]$ and $[Cu(PPh_3)_3I]$

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 74 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript

0.2%(w/v) solution of NaOH is measured to have resistivity 870.0 mΩ m. The molar conductivity of the solution will be_____ $\times 10^2 mSdm^2\ mol^{-1}$. (Nearest integer)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText

Question Number : 75 Question Type : SA Display Question Number : Yes Keyboard Layout : Inscript



Consider the above sequence of reactions. 151 g of 2-bromopentane is made to react. Yield of major product P is 80% whereas Q is 100%.

Mass of product Q obtained is_____ g.

(Given molar mass in $gmol^{-1}$ H : 1, C : 12, O : 16, Br : 80)

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText